Cardinal no. of Sets

If A, B, C are finite Sets and ∪ be the finite Universal Set then

i) $n(A \cup B) \Rightarrow n(A) + n(B) - n(A \cap B)$

Ex In a group of 65 people, 40 Like Cricket, 10 Like both Cricket and tennis. The no. of person Liking tennis only and not Cricket is.

Given

→ $65 \Rightarrow A \cup B$

$n(A) \Rightarrow 40$

$n(A \cap B) \Rightarrow 10$

$n(B) \Rightarrow ?$

$B \Rightarrow$ tennis

⇒ $n(A \cup B) \Rightarrow n(A) + n(B) - n(A \cap B)$

$65 \Leftrightarrow 40 + n(B) - 10$

$65 - 30 \Rightarrow n(B)$

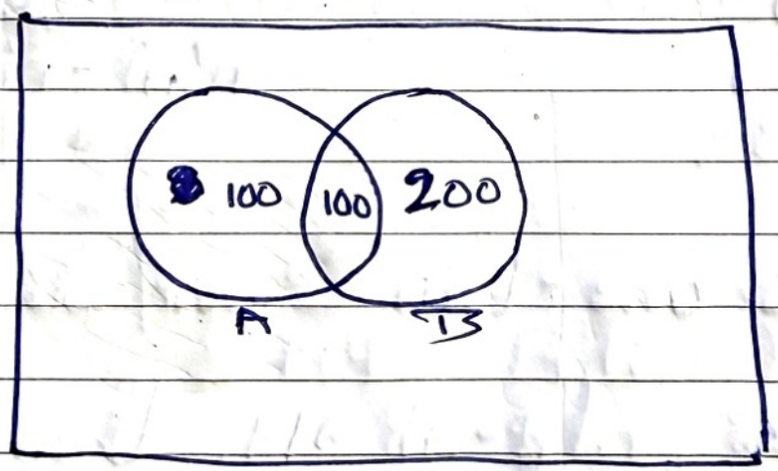
$n(B) \Rightarrow 35$

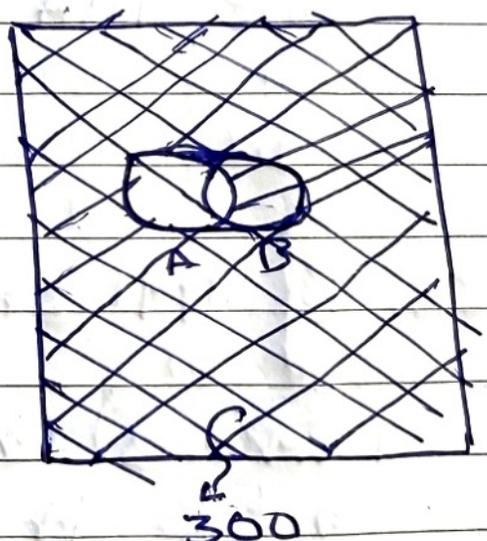
⇒ Liking tennis $35 - 10 \Rightarrow 25$

Ex. Let X be the Universal Set for Sets A and B. If $n(A) \Rightarrow 200$, $n(B) \Rightarrow 300$ and $n(A \cap B) = 100$, then $n(A' \cap B')$ in equal to 300 provided. $n(x)$ in equal to

⇒ Given,
X in an universal set
$n(A) \Rightarrow 200$
$n(B) \Rightarrow 300$
$n(A \cap B) \Rightarrow 100$
$n(A' \cap B') \Rightarrow 300$

⇒

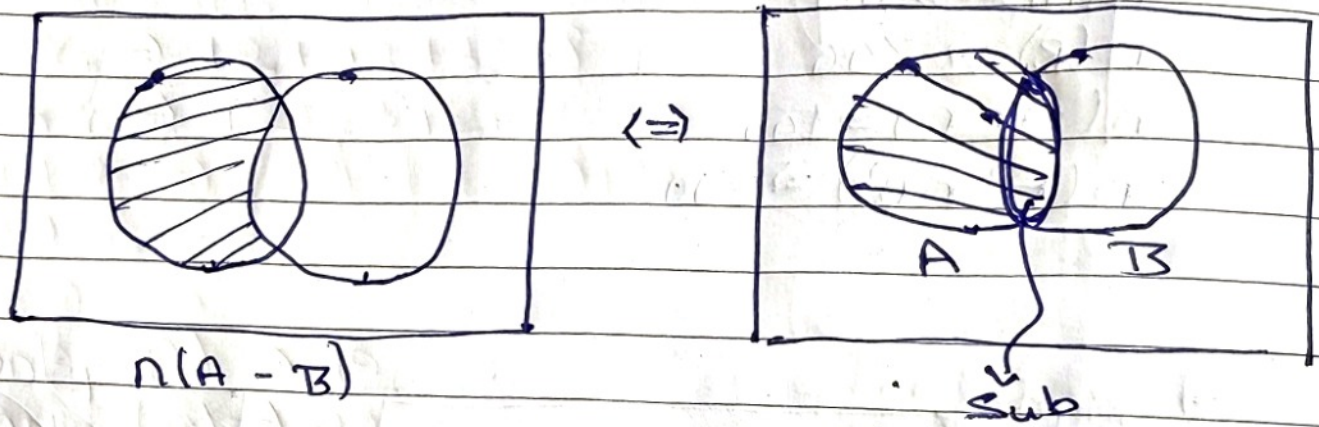




→ $n(A) + n(B) + n(A \cap B) + n(A' \cap B')$
$100 + 200 + 100 + 300 \Rightarrow 700$

2) If A, B, C are finite Sets and U be the finite universal Set then

* ⇒ $n(A - B) = n(A) - n(A \cap B)$



Ex. If A and B are two Sets Such that $n(A - B) \Rightarrow 24$, $n(B - A) \Rightarrow 19$, then $n(A \cap B) \Rightarrow 11$ then $n(A \cup B) \Rightarrow ?$

→ Given,

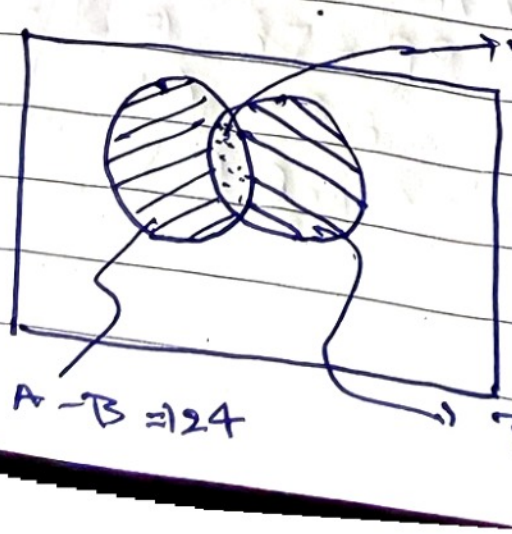
$n(A - B) \Rightarrow 24$

$n(B - A) \Rightarrow 19$

$n(A \cap B) \Rightarrow 11$

$n(A \cup B) \Rightarrow ?$

1)



$(A \cup B) \Rightarrow 11 + 24 + 19$

$\Rightarrow 54$

3) If A, B and C are finite Sets and U be the finite Universal Set then.

$n(A \cup B \cup C) \Rightarrow$

$n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

Ex. In a City, three daily newspaper A, B, C are published. 42 % of the people in that City read A, 51 % read B and 68 % C. 30 % read A and B; 28 % read B and C; 36 % read A and C; 8 % do not read any of the three newspaper, The percentage of person who read all the three papers is.

⇒ Given,

A ⇒ 42 %

B ⇒ 51 %

C ⇒ 68 %

$(A \cap B)$ ⇒ 30 %

$(B \cap C)$ ⇒ 28 %

$(A \cap C)$ ⇒ 36 %

8 % ⇒ U

⇒ M-1

$n(A \cup B \cup C) \Rightarrow n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

$\Rightarrow n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$

$\Rightarrow 42 + 51 + 68 - 30 - 28 - 36 + x$

$\Rightarrow 12 + 23 + 32 - x$

$-x \Rightarrow 12 + 23 + 32 - 8$

$-x \Rightarrow 12 + 23 + 24$

$x \Rightarrow 59$

Only (A) $\Rightarrow 12$

Only (B) $\Rightarrow 23$

Only (C) $\Rightarrow 32$

$8 \Rightarrow 12 + 23 + 32 - 30 - 28 - 36 + x$

$8 \Rightarrow 2 + (-5) + (-24) + x$

$-x \Rightarrow 2 - 5 - 24 - 8$

$-x \Rightarrow -3 - 24 - 8$

$-x \Rightarrow -35$

$x \Rightarrow 35\%$

$\Rightarrow n(A \cup B \cup C) \Rightarrow 92$

$92 \Rightarrow 42 + 51 + 68 - 30 - 28 - 36 + x$

$-x \Rightarrow 67 - 92$

$-x \Rightarrow -35$

$x \Rightarrow 35$

4) If A, B, C are finite Sets and U be the finite Universal Set then.

⇒ The no. of Elements in Exactly two of the Sets A, B, C

$$n(A \cap B) + n(B \cap C) + n(C \cap A) - 3n(A \cap B \cap C)$$

5) Number of Elements in at least two of the Sets A, B, C

$$\Rightarrow n(A \cap B) + n(B \cap C) + n(C \cap A) - 2n(A \cap B \cap C)$$

6) No. of Elements in Exactly one of the Sets A, B, C

$$\Rightarrow n(A) + n(B) + n(C) - 2n(A \cap B) - 2n(B \cap C) - 2n(A \cap C) + 3n(A \cap B \cap C)$$

Ex: A Class has 175 Students. The following data shows the no. of Students option one or more Subjects: Mathematics 100, Physics 70, Chemistry 40, Mathematics and physics 30, Mathematics and chemistry 28, physics and Chemistry 23, Math, physics and Chemistry 18. How many Students have opted Mathematics alone?

⇒ Given,

Total no. of students ⇒ 175

M ⇒ 100

P ⇒ 70

C ⇒ 40

M ∪ P ⇒ 30

M ∪ C ⇒ 28

P ∪ C ⇒ 23

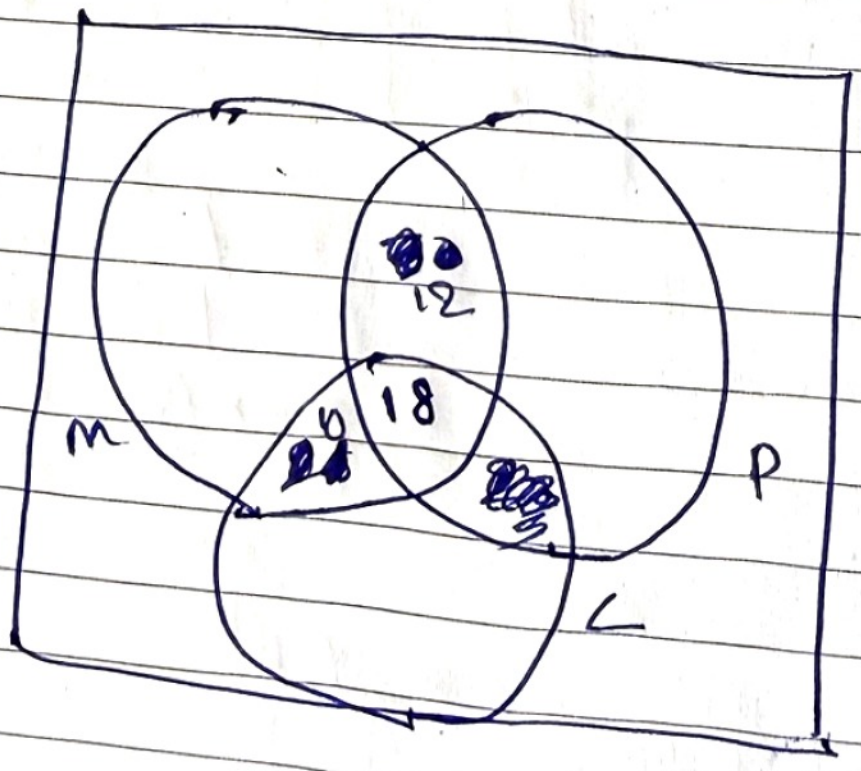
M ∪ P ∪ C ⇒ 18

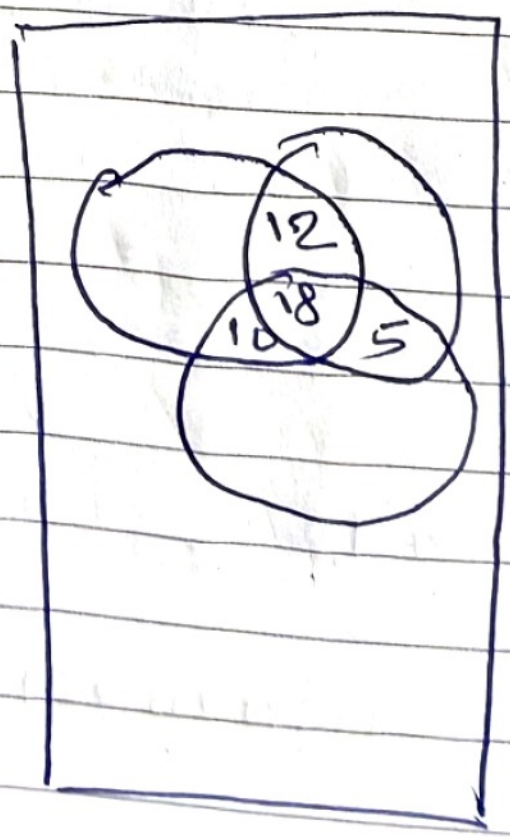
n(M) ⇒ ?

Rough









100 - 12 - 18 - 10 ⇒ 60

Ex) Out of 800 boys in the School, 224 played Cricket, 240 played hockey and 336 played basketball. of the total, 64 played both basket-ball and hockey; 80 played Cricket and basketball and 40 played Cricket and hockey; 24 played all the three games. the no. of boys who did not play any game is.

⇒ Given,

Total no. of boys ⇒ 800

C ⇒ 224

H ⇒ 240

B ⇒ 336

BUH ⇒ 64

CUB ⇒ 80

CUH ⇒ 40

BUCUH ⇒ 24

Did not play any games.

Rough

~~224~~ 96 = 128

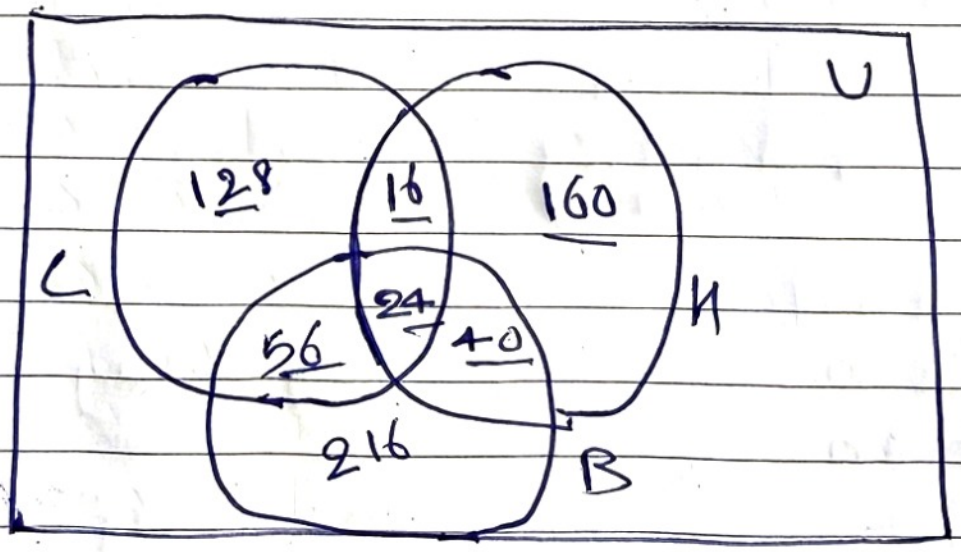
~~240~~ 80 = 160

336
120
216

128
116
144
160
300

56
64
120

800
216
184



U ⇒ did not play any games

⇒ 800 - 300 - 24 - 40 - 56 - 216

⇒ ~~800 - 120~~ ⇒ 160

⇒ ~~216~~

⇒ ~~184~~